ODS by mps

1. Graphical Method

- Use: Small-scale LP problems with 2 variables.
- Key Idea: Plot feasible region and slide objective line $z = ax + by$ to find optimum.
- Condition: Optimal at a vertex of feasible region.

2. Convexity & Concavity

- Convex Set:
  $(1-\lambda)x + \lambda y \in S$ for all $x, y \in S, \lambda \in [0,1]$
- Convex Function:
  $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$
- Concave Function:
  Reverse inequality above.

3. Hessian Matrix & Optimality

- Hessian: $H = \nabla^2 f(x)$
- Conditions:
  - FONC: $\nabla f(x^*) = 0$
  - SONC: $\nabla^2 f(x^*) \succeq 0$
  - SOSC: $\nabla^2 f(x^*) \succ 0$ ⇒ strict local min
- Global Optimum: If function is convex & feasible set convex.

4. Line Search Methods

- Golden Section: Uses ratio $\phi \approx 0.618$
- Fibonacci Search: Uses Fibonacci sequence for bracketing.
- Condition: Unimodal function over interval.

5. Newton's Method

- Update Rule:
  $x_{k+1} = x_k - [\nabla^2 f(x_k)]^{-1} \nabla f(x_k)$
- Condition: Hessian must be positive definite.

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6. Steepest Descent

- Update Rule:
  $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$
- Condition: $\nabla f(x_k)^T d_k < 0$ for descent.

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7. Quasi-Newton (DFP)

- Update:
  $x_{k+1} = x_k + \alpha_k d_k, \quad d_k = -B_k \nabla f(x_k)$
- Hessian Approximation: $B_{k+1} = B_k + \frac{ss^T}{s^T y} - \frac{B_k yy^T B_k}{y^T B_k y}$

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8. Conjugate Gradient Method

- Problem: Minimize $f(x) = \frac{1}{2} x^T Ax - b^T x$
- Conditions: A symmetric positive definite
- Update:
  - $r_k = b - Ax_k$
  - $d_k = r_k + \beta_k d_{k-1}$

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9. Least Squares

- Form: $Ax = b$, with $m > n$
- Solution:
  $x^* = (A^T A)^{-1} A^T b$
- Condition: $A^T A$ invertible.

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10. Simplex Method

- Conditions:
  - Convert inequalities using slack/surplus/artificial vars.

- Optimality: All $c_j - z_j \le 0$ (maximization).
- Big M: Add penalty $-MA$ in objective for artificial vars.

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11. Lagrange Multipliers

- Objective: Minimize $f(x)$ subject to $g_i(x) = 0$
- Lagrangian: $L(x, \lambda) = f(x) + \sum \lambda_i g_i(x)$
- Condition: $\nabla_x L = 0, \quad g_i(x) = 0$

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12. KKT Conditions

- For constraints $g_i(x) \le 0,\ h_j(x) = 0$
- Conditions:
  - Stationarity: $\nabla f + \sum \lambda_i \nabla g_i + \sum \mu_j \nabla h_j = 0$
  - Primal Feasibility: $g_i \le 0,\ h_j = 0$
  - Dual Feasibility: $\lambda_i \ge 0$
  - Complementary Slackness: $\lambda_i g_i(x) = 0$

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13. Penalty & Barrier Methods

- Penalty: Add $r \sum c_i^2(x)$ to objective; $r \to \infty$
- Barrier: Add $-\mu \sum \log(-g_i(x))$; $\mu \to 0$
- Goal: Convert constrained → unconstrained optimization.

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14. Frank-Wolfe Algorithm

- Step:
  - Solve: $s_k = \arg\min_{s \in D} \nabla f(x_k)^T s$
  - Update: $x_{k+1} = x_k + \gamma_k (s_k - x_k)$
- Condition: Function must be convex over convex set D
